Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

THE EVOLUTION OF MACHINE LEARNING AND AI

Week 5 Core Booklet

Teaching a Machine a Law of Nature: the required reading, in plain English

Module	The Evolution of Machine Learning and AI (5 ECTS, Semester 1)
Week	Week 5 of 12
Learning outcome	LO2: Characterise the training of the main families of model, here models that learn from a law rather than from data
Assessed in	Week 6 quiz (50% of the module mark). **Everything the quiz can ask about Week 5 is in this booklet.**
Reading time	About 45 minutes. Read it once, slowly, after the lecture and before the lab.
Goes with	The Week 5 lecture · the six-exercise lab, one exercise per chapter · `SUBMISSION.md`

What this booklet is. The **required** reading for Week 5, short on purpose. Same six parts as the lecture, same order, same words. Each chapter opens with its words in plain English. Every red ⋆ **For the Quiz** box is a statement the Week 6 quiz can test. Grey **Extra** boxes say where the deeper account lives in the **detailed** booklet, which you are not required to read. One equation appears, and it is read in words first; you will never solve it.

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YOUR WEEK, IN ORDER

1	The lecture (or its recording).	2 h
2	This booklet. Once, slowly.	45 min
3	The lab: six exercises, one per chapter of this booklet, in the same order. In each you predict first, run, and explain in your own words; in two you write the code. Some cells take a minute; the page says which.	2 h 35 min
4	Upload `SUBMISSION.md` (blank to start, filled in by you as you go), then compare it with the Week 5 Reference Answers on Tutors.	10 min
+	Optional: the three browser toys, the “break it” challenges, the detailed booklet.	only if you want

About five and a half hours. If an exercise takes more than 40 minutes, stop and ask in the channel.

1 A law is not a pattern

COVERS	Lecture Part 1 · Lab Exercise 1
TIME	7 minutes
IN BRIEF	A pattern in data is a nudge. A law of nature is a wall: exactly true, everywhere, whether or not you have data. Tell a model the law and it can learn with no data at all.

WORDS IN THIS CHAPTER, IN PLAIN ENGLISH

a pattern	something usually true in the data you happen to have, like "nearby pixels belong together"
a law	something always true, exactly, whether or not you have any data
a prior, a guess	what a model is told before it sees data; Week 4's inductive bias
a constraint	a guess so firm it removes answers entirely, instead of nudging away from them
error on the truth	how far the model's answer is from the real one, on average; smaller is better
the data regime	how much data you have; the lever that decides which methods work

1.1 The one picture: marking homework without the answer sheet

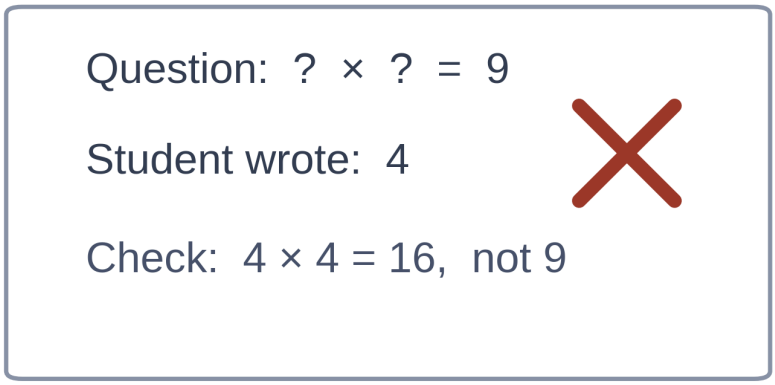


Figure 1: The homework is marked wrong without the answer sheet: put the answer back into the question.

You are marking a maths test and you have lost the answer sheet. The question: find the number whose square is 9. A student wrote 4. Can you mark it? Yes. Put 4 back into the question: $4 \times 4 = 16$, and 16 is not 9. Wrong. You never needed the answer sheet.

That is the whole of this week. **An equation is a question you can put an answer back into.** A model's guess can be checked against an equation anywhere you like, even where nobody has ever taken a measurement, because checking does not require knowing the answer. And it connects to Week 3 in one line: the loop was guess, check, blame, nudge, and this week only the **check** changes. Instead of "how far from the measurement?" it asks "how far from obeying the law?".

1.2 Two kinds of advice

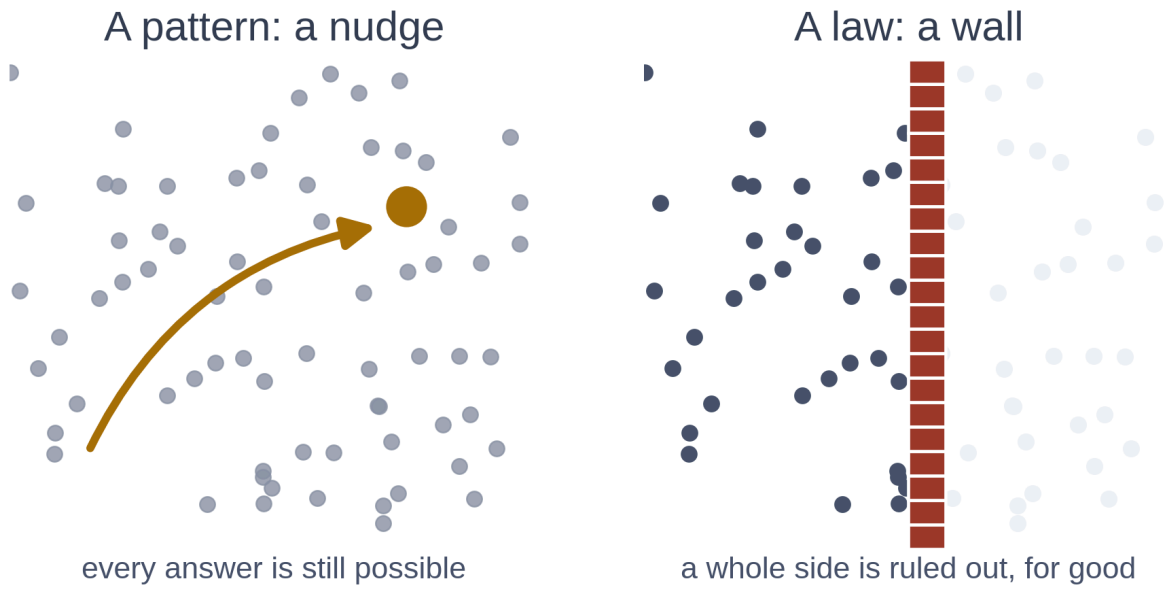


Figure 2: Left: a pattern nudges towards a likely answer, and every answer stays possible. Right: a law is a wall; a whole side of answers is gone for good.

EVERYDAY ANALOGY

"People in high-visibility jackets are often construction workers." Useful, usually right, sometimes wrong. It rules out nothing; you still have to look at every case. That is a **nudge**.

"This person is standing in Dublin, so they are not in Cork." Not a hint. A whole set of answers is gone for good, and no new evidence about the jacket brings it back. That is a **wall**.

A pattern gives a model a nudge towards answers that have usually been right. A law gives it a wall: a region of answers that is simply excluded. Week 4 gave models nudges. This week gives them walls.

1.3 The one equation of the week, in words

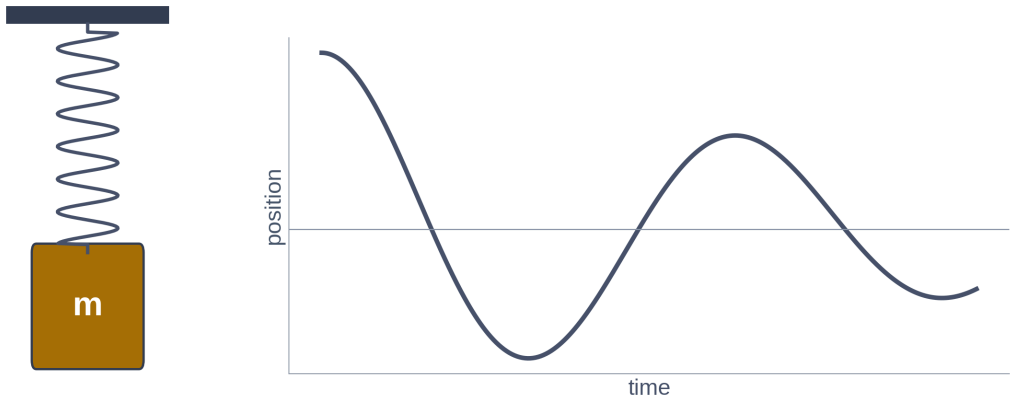


Figure 3: A weight on a spring: pull it down, let go, and it bobs up and down a little less each time until it settles.

Hang a weight on a spring, pull it down, let go. It bobs up and down, a little less each time, until it stops. Three things decide how it moves: how heavy the weight is (m), how much friction slows it (c), and how stiff the spring is (k). The rule it obeys, in words: **the weight times how fast the speed is changing, plus the friction times the speed, plus the stiffness times the position, always adds up to exactly zero.** In symbols, so you recognise it:

$$mu'' + cu' + ku = 0$$

u is the position, u' is the speed (how fast the position is changing) and u'' is how fast the speed is changing. You will never solve this. You only need one fact about it: it is a question you can put an answer back into. Give any guess for the position over time, and the equation tells you how badly the guess breaks the rule at any moment, without knowing the right answer.

1.4 The result that sets up the week

Same network. Same number of dials. Same optimiser. Same number of trips round the loop. The only difference is what the check compares the guess against.

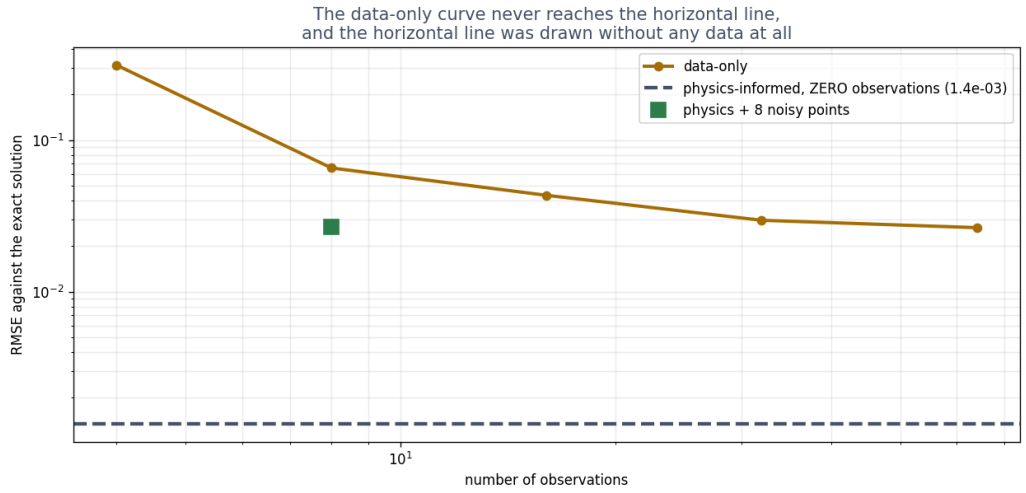


Figure 4: The data-only model improves with measurements and flattens out at about 0.026, the size of the noise. The physics-informed model, using zero measurements, scores 0.00135: twenty times better, and it cannot be plotted, because its position on the horizontal axis is zero.

model	measurements it saw	error on the truth
data-only	64	0.026
physics-informed	**0**	**0.00135**

The physics-informed model never saw a single measurement of the thing it was predicting. It was told what the answer must **satisfy** instead of being shown examples of it.

1.5 Why this matters where you will work

Week 4 ended with a Vision Transformer beating a stencil model given 300 million pictures, and losing given 1.3 million. You will not have 1.3 million labelled examples. You will have forty readings from one wind turbine, or two hundred patients from one hospital. In that world **more data is not available at any price**. The only lever left is what you already know, and engineering and science come with four centuries of extremely reliable knowledge that machine learning, until recently, made almost no use of.

★ FOR THE QUIZ

- A **pattern** is a nudge: usually right, eliminates nothing. A **law** is a wall: exactly true everywhere, and it removes whole sets of answers.
- An equation is a question you can put an answer back into. Checking a guess against it needs no answer sheet.
- Same network, same optimiser, different check: with **zero** measurements the physics-informed model scored 0.00135; with 64 measurements the data-only model scored 0.026.
- Where you will work, more data is not for sale, so what you already know is the only lever.

EXTRA · OPTIONAL · NOT ON THE QUIZ

Detailed booklet Chapters 1 to 5: the formal statement of a prior against a constraint, reading a differential equation, and where scientific knowledge lives.

2 Checking the equation where there is no data

COVERS Lecture Part 2 · Lab Exercise 2
TIME 9 minutes
IN BRIEF The method in plain steps: put the guess back into the equation, square what is left over, and treat that as the loss. Only the check step changed. The one new trick is taking the slope of a slope, exactly.

WORDS IN THIS CHAPTER, IN PLAIN ENGLISH

residual	how far a guess is from obeying the equation; zero means it obeys
physics loss	the residual, squared and averaged over many places; the new check step
collocation points	the places where the equation is checked; free, because no measurement is needed
derivative, slope	how fast something is changing; u′ is the speed, u″ is how fast the speed changes
automatic differentiation	the computer working out slopes exactly, by tracking every arithmetic step
physics-informed network (PINN)	an ordinary network whose check is an equation; not a new kind of network

2.1 The whole method, before any notation

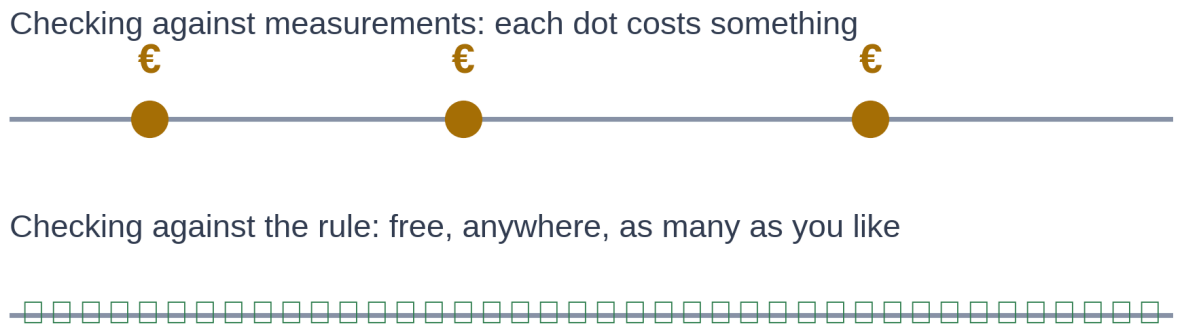


Figure 5: Measurements cost something, so you have few. Checks against the rule are free, so you can have as many as you like, anywhere.

Ordinary supervised learning compares the model against measurements, at the places you have measurements. Each dot cost somebody a sensor, an experiment, an afternoon. **Physics-informed learning** compares the model against an equation, at places where you have nothing at all. Each dot is free, because checking an equation does not require knowing the answer. So you can put a thousand checkpoints anywhere you like, including places where measuring is impossible.

2.2 What the check actually computes

1. Start with a time. There is no measurement attached to it, and there does not need to be.
2. The network produces its current guess for the position at that time.
3. Work out the speed, and how fast the speed is changing, from the network's own guess: the slope of the network's output with respect to its input, and the slope of that. This is the operation Weeks 3 and 4 never needed.

4. Put position, speed and change-of-speed into the equation. If the guess were the true motion, the sum would be exactly zero. Whatever it is instead is the **residual**, the leftover.

Nowhere in those four steps does the true answer appear. Square the residual, average it over a few hundred times, and that is the entire loss. Then blame and nudge, exactly as in Week 3.

EVERYDAY ANALOGY

Try it with a pencil. Suppose the spring has $m = 1$, $c = 0$ (no friction) and $k = 1$, and a student guesses that the position is always 0.7. Then the speed is 0 and the change of speed is 0, so the equation's left side is $1 \times 0 + 0 \times 0 + 1 \times 0.7 = 0.7$. The leftover is 0.7, not 0, so the guess is wrong, and you found that out without knowing the right answer. The lab prints this exact check for three candidate answers.

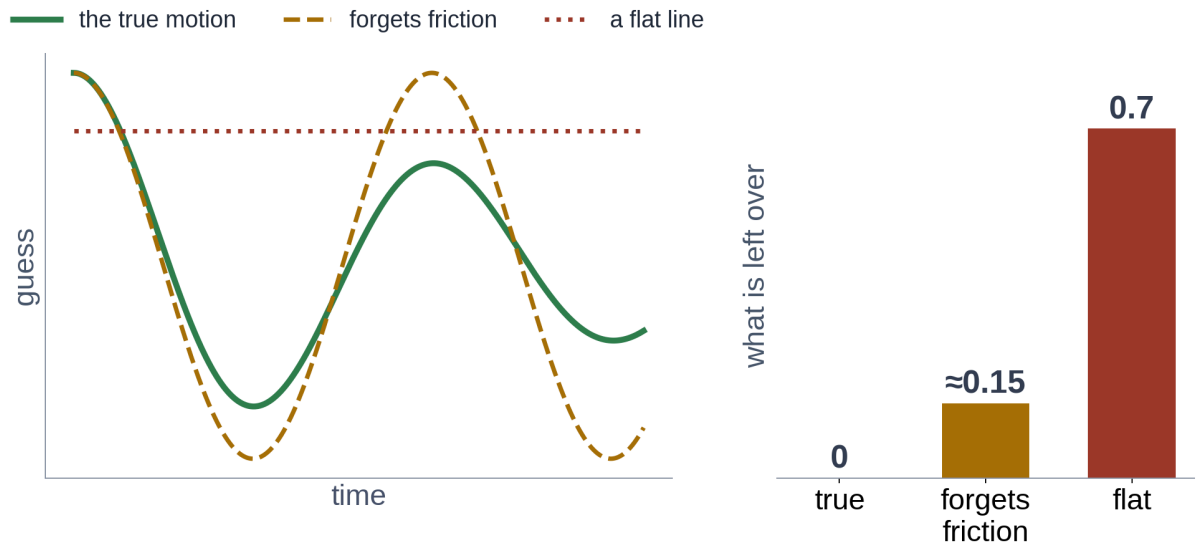


Figure 6: Three guesses put back into the rule. The true motion leaves nothing over; forgetting friction leaves a little; a flat line leaves a lot. The rule did the marking.

2.3 Only one thing changed

the loop's step	Weeks 3 and 4	this week
the network	an ordinary network	the same
guess	input in, answer out	the same
check	distance from the measurement	**how far from obeying the equation**
blame	which dials caused it	the same
nudge	turn the dials a little	the same

IN PLAIN ENGLISH

A physics-informed network is **not a new kind of network**. There is no special layer and no special architecture. It is an ordinary network with an unusual check. If you go looking for "the PINN architecture" you will be confused, because there is none.

2.4 The one genuinely new trick

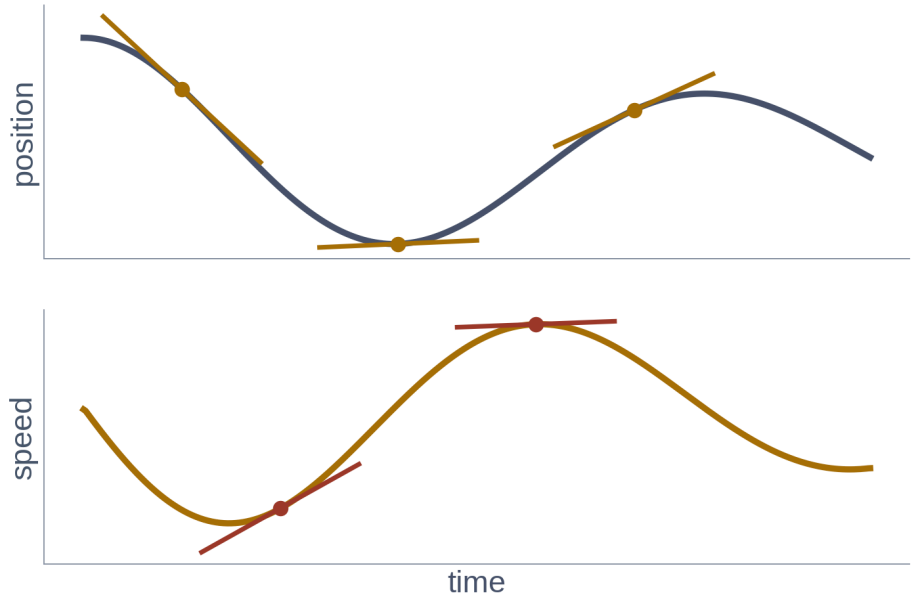


Figure 7: Top: the slope of the position is the speed. Bottom: the slope of the speed is how fast the speed is changing. The rule needs both: the slope of a slope.

The check needs the slope, and the slope of the slope. Then, to blame and nudge, the computer needs the slope of **that** with respect to every dial. Ordinary training software works out slopes as plain numbers, and you cannot take the slope of a number. The trick is to make the computer **record how each slope was worked out**, so that the slope is itself something you can take the slope of. The lab's own 250-line engine does this; checked against slopes worked out by hand up to the third slope-of-a-slope, its worst error is **1.22e-16**, the rounding error of the computer's arithmetic. In PyTorch this is what `create_graph=True` means.

COMMON MISTAKE

Estimating the slope by nudging the input a tiny step and measuring the change. It works for a first slope and fails for the slope of a slope: a tiny difference divided by a tiny number squared is swamped by rounding. Measured in the lab: the exact engine gives -0.03664877 for a second derivative and the nudging method -0.03664985; the disagreement is the nudging method's error.

★ FOR THE QUIZ

- A **residual** is how far a guess is from obeying the equation: put the guess back into the question and see what is left over. Zero means it obeys.
- A **physics loss** is the residual, squared and averaged at many **collocation points**, places where the equation is checked and no measurement is needed. No answer appears anywhere in it.
- A physics-informed network is an ordinary network with a different check; guess, blame and nudge are Week 3′s, unchanged.
- The one new trick: the computer must be able to take the **slope of a slope**, exactly. Nudging and measuring is not good enough.

EXTRA · OPTIONAL · NOT ON THE QUIZ

Detailed booklet Chapters 6 to 9: the engine's design, differentiating a derivative, and the PINN end to end.

3 What the law buys

COVERS Lecture Part 3 · Lab Exercise 3
TIME 7 minutes
IN BRIEF A model that knows the law keeps working beyond the data, because the law was known there too. And it inherits every consequence of the law, such as conservation of energy, whether or not you thought to check.

WORDS IN THIS CHAPTER, IN PLAIN ENGLISH

extrapolation	answering outside the range where you had any measurements
degradation	how many times worse a model gets outside its data than inside; 1.0 means no worse
energy	for the spring: one number made from the speed and the position; you only need to know it is one number
conservation	a quantity that stays exactly the same over time when nothing is added or lost
drift	how far a model's version of a conserved quantity wanders from constant
a consequence	something the equation implies but nobody wrote into the loss

3.1 Beyond the data

Both models get measurements on the first 40% of the time only. Both are scored on the rest.

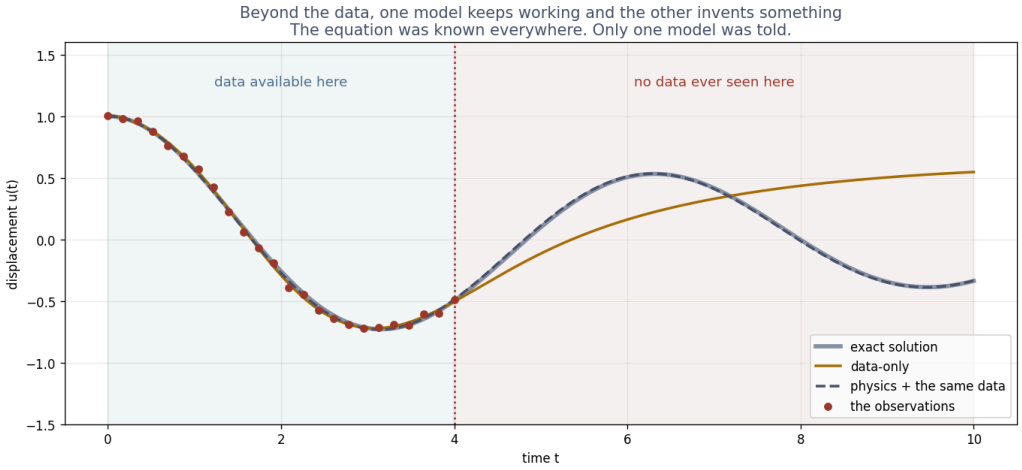


Figure 8: Data on the left of the dotted line only. The data-only model does something smooth and arbitrary beyond it, and does not warn you. The physics-informed model follows the truth, because the equation was known out there too.

model	error where data existed	error beyond the data	how many times worse
data-only	0.018	0.500	**27.6**
physics + the same data	0.0086	0.0087	**1.0**

EVERYDAY ANALOGY

You are asked where a thrown ball will land, having watched only the first half-second of its flight. A model fitted to what you saw describes that half-second beautifully and has no opinion about the rest; stretched further, it does something smooth and arbitrary, and it does not warn you. A model that knows the ball is under gravity can tell you where it lands, because gravity applies for the whole flight. The second model is not stretching the data. It is following the law.

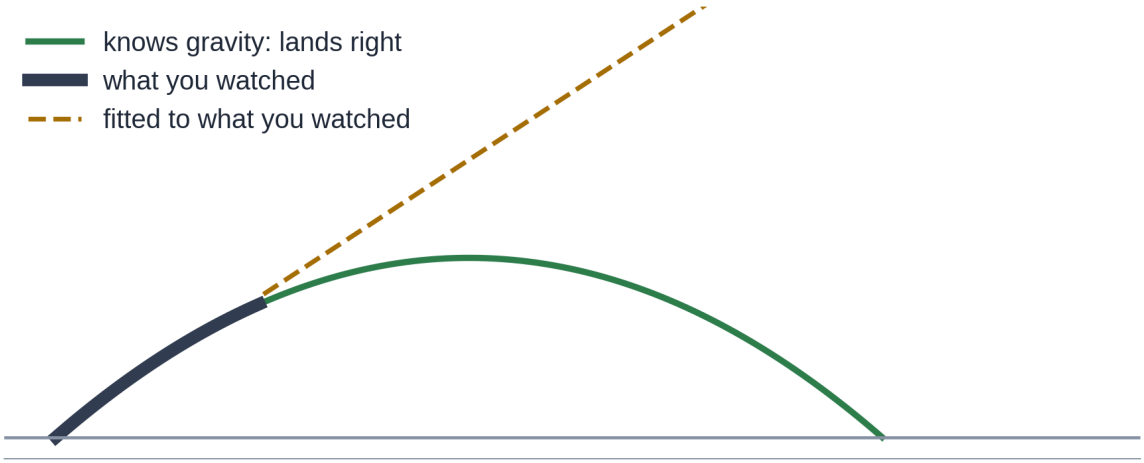


Figure 9: Watch only the start of a throw (thick line). Copying the shape of what you saw keeps going up (dashed); knowing gravity brings the ball down in the right place (green).

3.2 Something nobody put in the loss

Take the friction away. Now the spring bobs for ever and its energy stays exactly the same the whole time. **Nobody tells either model this.** It is not in the loss, not in the network, not in the data.

model	how far its energy drifted	what that means
data-only	**16.3%**	physically impossible, and nothing flagged it
physics + noisy data	5.1%	better
physics only, no data	**0.2%**	kept constant a thing it was never told about

A model that obeys the equation inherits **everything the equation implies**, whether or not you thought to check it. A model that fits points inherits agreement with those points, and nothing else.

EVERYDAY ANALOGY

Two students. One has memorised the worked examples; the other understands the subject. Both answer the questions they have seen. You find out which you have by asking a question that was not on the practice sheet. Energy drift is that question, and it costs nothing to ask. Your test set covers the things you thought to measure; the law covers the things you did not.

★ FOR THE QUIZ

- **Beyond the data**: given measurements on the first 40% only, a data-only model got 27.6 times worse outside them; the physics-informed model got no worse (1.0), because the law was true out there too.
- A data-only model does not know that it does not know. It fails smoothly and silently outside its data.
- **Conservation**: a frictionless spring keeps its energy exactly. Nobody told either model. Data-only drifted 16.3%; physics-only drifted 0.2%.
- A model that obeys the equation inherits everything the equation implies. Checking a consequence you never trained on is how you tell a memoriser from an understander.

EXTRA · OPTIONAL · NOT ON THE QUIZ

Detailed booklet Chapters 10 to 14: noise robustness, the heat equation (a two-input version), and the full write-ups.

4 What it costs, and the two ways it goes quietly wrong

COVERS Lecture Part 4 · Lab Exercise 4
TIME 9 minutes
IN BRIEF A soft penalty needs a weight nobody can derive; a hard constraint builds the condition into the arithmetic. A small loss only speaks for the places you checked. And a wrong law is worse than no law: the loss stays small while the answer goes wrong.

WORDS IN THIS CHAPTER, IN PLAIN ENGLISH

initial condition	where the spring starts and how fast it is moving at time zero
soft penalty	adding "and please satisfy the start" to the loss, with a weight saying how much it matters
hard constraint	writing the model so the start holds automatically, whatever the dials do
the weight, λ	the number that says how much one part of the loss matters compared with another; nobody can derive it
wrong physics	being told an equation with the wrong number in it, and trusting it
a dashboard	the training charts you watch; here, the ones that fail to go red

4.1 The weight nobody can derive

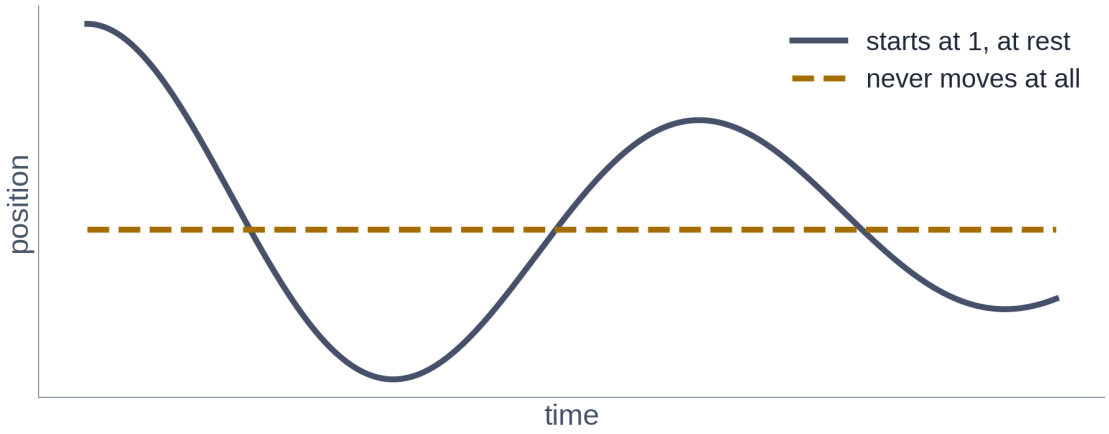


Figure 10: Both lines obey the spring's rule. Only the start tells them apart, so the start must be given too.

The equation on its own does not choose a motion: "the weight never moves" obeys it perfectly. What picks out **our** motion is where it starts: at position 1, at rest. The obvious way to tell the model is a **soft penalty**: loss = (how badly it breaks the equation) + λ × (how badly it misses the start), where λ is a number you choose.

how the start was imposed	start position (should be 1)	error on the truth
soft penalty, λ = 1	0.99988	0.00154
soft penalty, λ = 50	0.99941	**0.32225**
soft penalty, λ = 500	0.99953	**0.66444**
hard constraint	**1.00000**	**0.00135**

A spread of over 400 times, caused entirely by one number that nobody can work out in advance. Now read the middle column: **every version got the start right to four decimals**. The bad runs are not breaking the starting condition. They are obeying it while neglecting the equation.



Figure 11: Badly weighted instructions: a perfect red door on a house that is not standing straight.

EVERYDAY ANALOGY

Two instructions to a builder: make the house stand up, and paint the front door red. Tell them the red door is five hundred times more important and you get a beautifully painted door on a house that is not quite standing up. The builder followed the instructions. The instructions were badly weighted.

4.2 The alternative: make it arithmetic

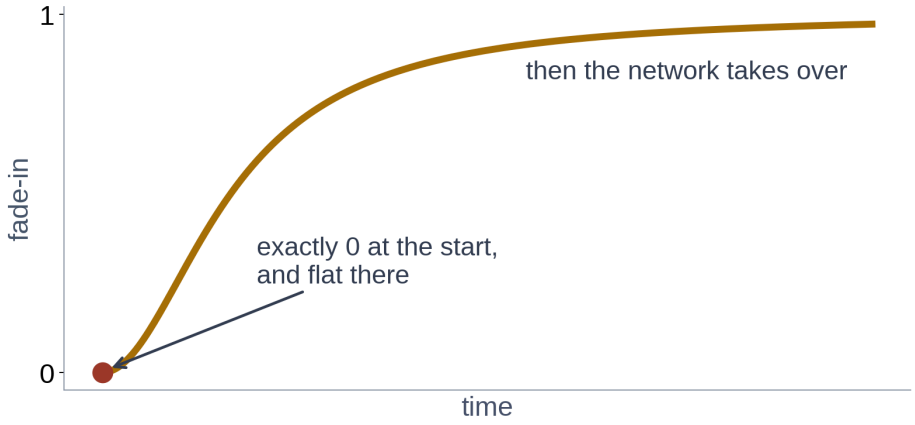


Figure 12: The fade-in is exactly zero at the start, and flat there, so the start is always exact whatever the network says.

Instead of asking the model to satisfy the start, make it impossible not to. The model's answer is: **the starting position, plus a fade-in factor, times whatever the network says**. The fade-in is zero at time zero, so the position is exactly the start whatever the dials are; and its slope is also zero at time zero, so the speed is exactly zero too. In symbols, $u(t) = 1 + s(t)N(t)$ with $s(t) = \frac{t^2}{1+t^2}$. Now the start holds automatically, at every step, whatever the optimiser does. The loss has one part, there is nothing to balance, and there is no λ.

MEASURED IN THE LAB

Done the same way on heat spreading along a rod (two inputs, position and time; the start and both ends built in): error **0.0004** using no data, against 0.018 for a data-only model given 128 measurements. About forty times better.

4.3 A small loss proves less than you think

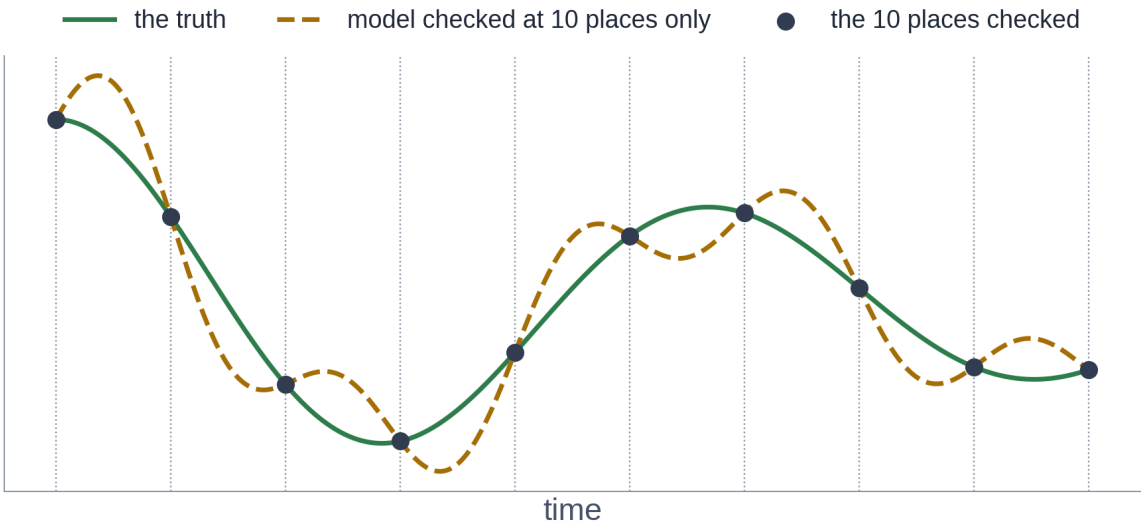


Figure 13: Checked at only ten places, the model obeys the rule at those places and wanders between them.

Checkpoints are free, which is the method's great advantage. It is also its characteristic failure.

places the equation is checked	loss the model reports	true error, everywhere
10	**0.0000026 (tiny!)**	**0.30**
50	0.000049	0.0015

With only ten checkpoints the reported loss is the **smallest**, because the equation really is obeyed at those ten places. The curve between them wanders freely. **The loss is not lying. It is answering a different question from the one you meant to ask.** Three cheap checks: check again at fresh places; add more places and see whether the answer moves; check a consequence, such as energy.

4.4 When the physics is wrong

The real spring has friction $c = 0.2$. Every model gets real measurements. The physics-informed ones are also told a value of c, and two are told the wrong one.

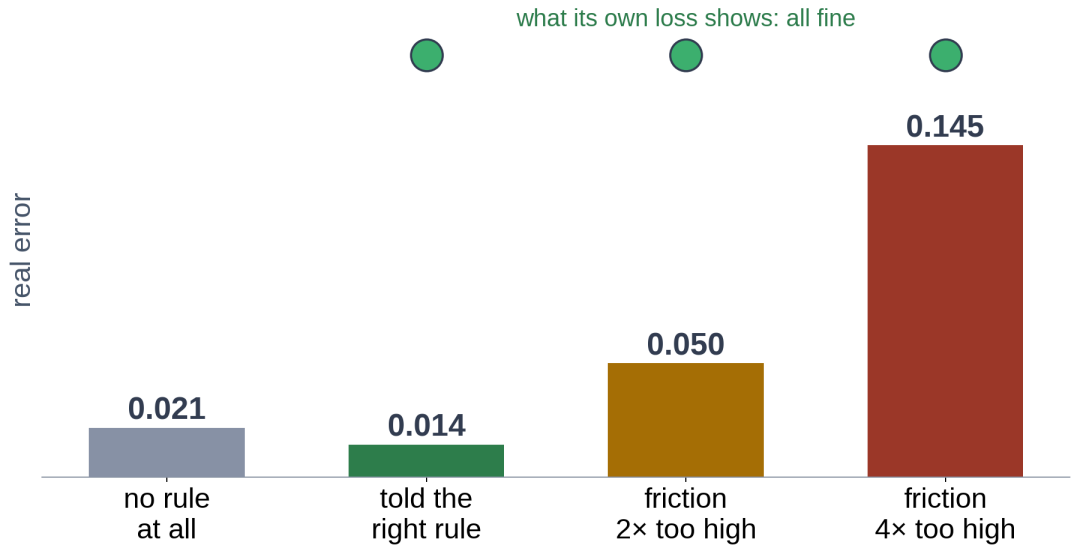


Figure 14: Real error for each model; the green lights are what each model's own score shows. Told friction four times too high, it is seven times worse than using no rule, and the score still says all is fine.

model	true error	the loss it reports
data-only, no physics	0.0215	n/a
physics, told c = 0.2 (correct)	**0.0142**	0.00058
physics, told c = 0.4 (twice too high)	0.0499	0.0047
physics, told c = 0.8 (four times too high)	**0.1455**	0.023

Told the friction is four times too high, the model is **seven times worse than assuming nothing at all**. And read the last column: the loss it reports stays small the whole way. Nothing on a training dashboard goes red. You cannot catch this by watching the loss. You catch it only by checking against the world.

EVERYDAY ANALOGY

A confident expert with an out-of-date fact is more dangerous than an honest beginner. The beginner says "I am not sure", so you check. The expert gives a precise, well-reasoned answer built on a wrong premise, and every calculation downstream inherits the error, because everything is consistent with everything else. A physics-informed model with the wrong equation is the confident expert.

⋆ FOR THE QUIZ

- The equation alone does not pick an answer; the **initial condition** does. A **soft penalty** adds it to the loss with a weight λ that nobody can derive: errors spread over 400 times across three values of λ, while every run got the start right to four decimals.
- A **hard constraint** writes the model so the start holds automatically (a fade-in that is zero at $t = 0$). No λ, nothing to balance.
- A small loss proves less than you think: it reports obedience at the places you checked and nothing about the places you did not. Check at fresh places; check a consequence.
- **A wrong law is worse than no law**: told friction four times too high, the model was seven times worse than assuming nothing, and its loss stayed small the whole way.

EXTRA · OPTIONAL · NOT ON THE QUIZ

Detailed booklet Chapters 15 to 17 and 22: loss balancing, curriculum training, and the honest counter-examples in full.

5 Four more ways to build in science

COVERS Lecture Part 5 · Lab Exercise 5
TIME 8 minutes
IN BRIEF Symmetry: feed the model only what the symmetry cannot change, and it cannot get it wrong; but a false symmetry is a wall. A neural ODE learns the rule, not the answer. A neural operator learns a whole family at once. Diffusion runs a known physical process backwards.

WORDS IN THIS CHAPTER, IN PLAIN ENGLISH

symmetry	something you can do to the input (turn it, shift it) that must not change the answer
invariant model	a model that reads only things the symmetry cannot change, such as distances
neural ODE	a model that learns the rule for the next small step instead of the whole answer, then rolls forward
neural operator	a model that learns to solve a whole family of equations at once, one answer per new setting
diffusion model	a picture generator that runs a known physical process (ink spreading) backwards
rollout	applying the next-step rule again and again, further than the training covered

5.1 Symmetry: the only guess that is exactly true

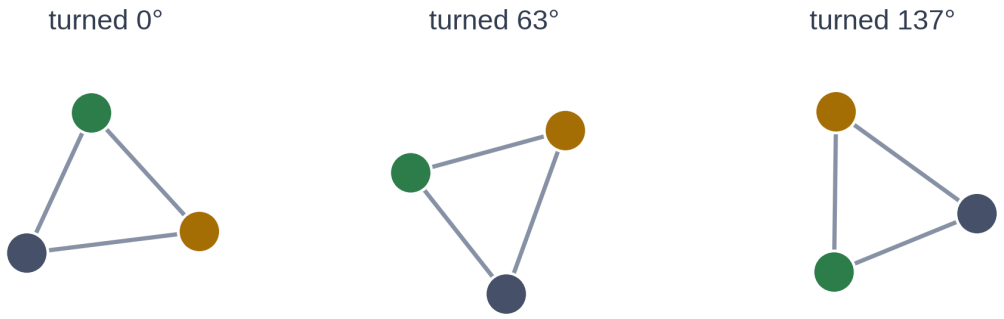


Figure 15: The same three particles turned three ways. The distances between them never change, so neither can the energy.

Three particles whose energy depends only on the **distances** between them. Turn the whole arrangement and the energy cannot change; that is a symmetry of the law itself. A model that reads raw positions must **learn** this from examples: all six numbers change when you turn the picture. A model that reads only the three distances **cannot get it wrong**, because a distance does not change when you turn the picture.

model	change in its answer when the input is turned	as % of the answer
reads raw positions	1.76e-01	**6.4%**
reads distances only	**8.88e-16**	**0.0%**

That second number is the rounding error of the computer's arithmetic: the same number computed twice. On orientations never seen in training, the raw-position model got 6.2 times worse; the distance model 1.0.

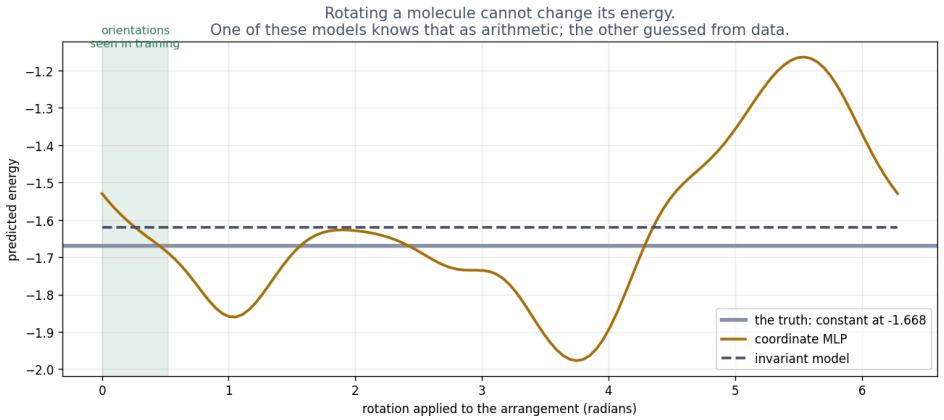


Figure 16: Turn the input through every angle. The raw-position model's answer wobbles; the distance model's does not move.

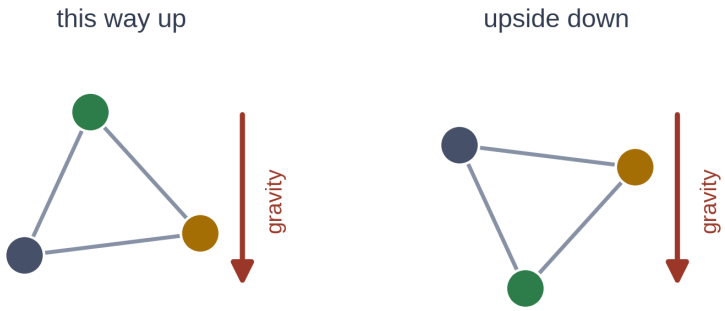


Figure 17: With gravity, which way up matters. A model built to read only distances cannot see this, at any amount of data.

And the honest half. Add gravity, so the energy genuinely does depend on which way up the arrangement sits. Now the raw-position model scores 0.11 and the distance model 0.24, and **cannot do better at any dataset size**, because it was built unable to see which way up anything is. A constraint is a promise about the world: true, an exact guarantee; false, a model that cannot represent the truth however much data you give it. Same lesson as Chapter 4, in a different costume.

5.2 Learn the rule, not the answer

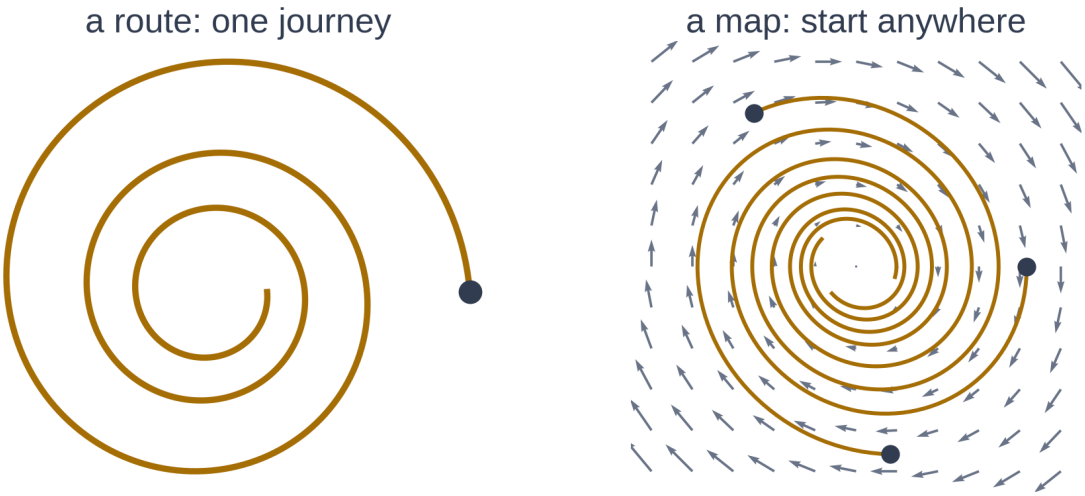


Figure 18: A route is one journey. A map, the little arrows, lets you start anywhere: that is learning the rule for the next step.

A physics-informed network learns one motion: the spring from this start. A new start means training again. A **neural ODE** learns the rule for the next small step ("given where you are and how fast you are going, here is where you will be a moment later"), and a stepper rolls it forward from wherever you like.

EVERYDAY ANALOGY

Memorising a route against learning to read a map. Memorise the route and you can make that one journey. Learn the map and you can start anywhere, because you have the rule rather than the answer.

MEASURED IN THE LAB

Trained on 20 steps, asked to carry on for 60: a model that memorised the motion got 11.6 times worse; the neural ODE 1.7; a neural ODE told the known half of the rule (speed is the slope of position) 1.6, with fewer dials.

5.3 Solve a whole family at once

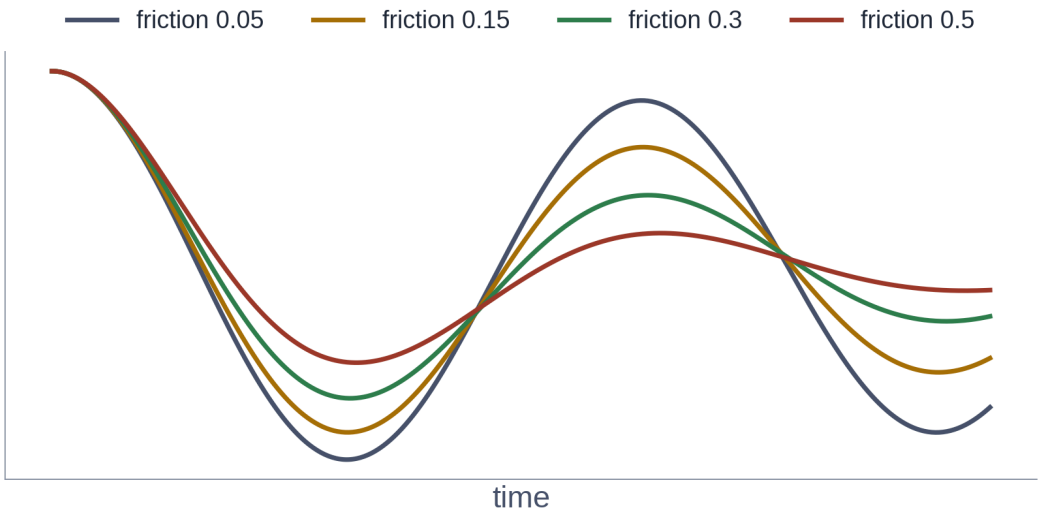


Figure 19: One family: the same spring rule with four different amounts of friction.

Designing a bridge or a wing means solving the same equation thousands of times with slightly different numbers in it. A **neural operator** learns the map from "the numbers in the equation" to "the whole answer" for a family at once: train once on forty solved cases, then each new case is one quick pass. A new friction value costs **a quarter of a millisecond instead of several seconds** to train a fresh model, and the answer is less accurate (in the lecture's full run about 60 times; in the lab's short run only slightly). Both facts are the point: it is a trade.

5.4 A picture generator that is a physics paper

Ink in water. Random jostling dissolves the structure, and after long enough the glass has forgotten where the drop was. That forward process is **given by physics, learned by nobody**. Each step added only a little noise, so each step is nearly reversible: learn that one small reversal, apply it many times, and you run the diffusion backwards from pure noise to a picture. The stability came from that structure, not from a cleverer optimiser. And the vocabulary (diffusion, equilibrium, annealing) was borrowed from a century of physics.

★ FOR THE QUIZ

- **Symmetry**: if turning the input must not change the answer, build a model that reads only what turning cannot change (distances). The guarantee is exact (8.88e-16). A **false symmetry is a wall**: the model cannot represent the truth at any dataset size.
- A **neural ODE** learns the rule for the next step, not the answer, so it can roll forward from any start (1.7 times worse at three times the length, against 11.6 for a memorised motion).
- A **neural operator** learns to solve a whole family of equations at once: a fraction of a millisecond per new case instead of several seconds, and less accurate. A trade.
- A **diffusion model** runs a known physical process (ink spreading) backwards, one small step at a time. Its stability comes from that structure.

EXTRA · OPTIONAL · NOT ON THE QUIZ

Detailed booklet Chapters 18 to 24: equivariance beyond rotation, structured dynamics, the operator landscape, and the score function.

6 Choosing

COVERS Lecture Part 6 · Lab Exercise 6
TIME 5 minutes
IN BRIEF Match the method to what you know, not to the method's fame. Physics-informed learning is not a better way to solve a standard equation. Seven questions before you build.

WORDS IN THIS CHAPTER, IN PLAIN ENGLISH

classical solver	sixty years of tried-and-tested software that solves standard equations exactly and fast
inverse problem	you have measurements and an equation with an unknown number in it; find the number
surrogate	a fast stand-in for a slow simulation
standard domain	a simple, regular shape (a rod, a square, a box) where classical solvers shine
mesh	the grid a classical solver needs; drawing one for an awkward shape can be the expensive part

6.1 Match the method to the knowledge

if you have	reach for	because
an equation, one situation	a physics-informed network	needs no data; gives the answer and its slopes
an equation, thousands of situations	a neural operator	train once, answer each new case in a millisecond
an equation on a simple, standard shape	**a classical solver**	faster, more accurate, with guarantees
an equation with an unknown number in it	a physics-informed network, inverse	make the unknown a trainable dial
half the rule known, half not	a structured neural ODE	impose the known half, learn the rest
an exact symmetry	a symmetry-aware model	exact guarantee, less data needed
no reliable physics	**ordinary machine learning**	a guessed law is worse than none

Reaching for the famous method when what you actually have is a symmetry is a common and expensive mistake.

6.2 The claim worth being careful about

IN PLAIN ENGLISH

Physics-informed machine learning is not a better way to solve differential equations. For a standard equation on a simple shape, sixty years of numerical analysis has produced solvers that are faster, more accurate, and come with guarantees. Use them. It is a better way to do several **other** things: combine a few messy measurements with a known law; find an unknown number in an equation from measurements; handle awkward shapes where drawing the grid is the expensive part; build a fast stand-in for a slow simulation. Being able to say this clearly is the mark of understanding the field rather than its marketing.

6.3 Seven questions for anything you build with a law in it

1. What do I know, said as a sentence a non-specialist would understand?
2. Is it exactly true, approximately true, or a guess?
3. Is it a hard constraint (built into the arithmetic) or a soft penalty (a weight in the loss)? They carry very different guarantees.
4. Where is the loss actually checked, and what does that leave unchecked?
5. What consequence can I check that I never trained on? Energy is the classic.
6. What happens if my physics is wrong by 20%?
7. Would a classical solver, or ordinary machine learning, have done this better?

★ FOR THE QUIZ

- Match the method to the knowledge: one situation, a PINN; thousands, an operator; a simple standard shape, a classical solver; an unknown number, an inverse problem; half a rule, a structured neural ODE; a symmetry, a symmetry-aware model; no reliable physics, ordinary machine learning.
- Physics-informed learning is **not a better way to solve equations**. It is a better way to combine sparse data with a known law, to infer unknown numbers, to handle awkward shapes, and to build fast stand-ins.
- Seven questions, starting with: what do I know, said plainly, and is it exactly true, roughly true, or a guess?

EXTRA · OPTIONAL · NOT ON THE QUIZ

Detailed booklet Chapters 25 to 27: case studies, and the responsibility angle (models that cannot lie about physics).

7 The whole week on one page

COVERS Revision
TIME 3 minutes to read; 10 to memorise
IN BRIEF Every quiz-able statement from Week 5, in one place.

Topic	The sentence to be able to say
Pattern against law	A pattern nudges; a law is a wall. A law is exactly true everywhere, with or without data.
The homework test	An equation is a question you can put an answer back into. Checking needs no answer sheet.
The result	Zero measurements: 0.00135. Sixty-four measurements, data-only: 0.026. Same network, different check.
Residual and physics loss	The leftover when the guess goes back into the equation, squared and averaged at free checkpoints. No answer appears.
Not a new network	Only the check changed. There is no PINN architecture.
The new trick	The slope of a slope, exactly; the engine is right to 1.22e-16. Nudging and measuring is not enough.
Beyond the data	Data on the first 40%: data-only 27.6 times worse outside; physics-informed 1.0.
A consequence for free	Energy in nobody's loss: data-only drifted 16.3%, physics-only 0.2%. A model that obeys the equation inherits what it implies.
Soft against hard	Soft penalties at λ = 1, 50, 500: 0.0015, 0.32, 0.66, all with the start right to four decimals. Hard constraint: 0.00135, no λ.
A small loss	It speaks only for the places you checked. Check fresh places; check a consequence.
A wrong law	Friction four times too high: seven times worse than no physics, and the loss stayed small.
Symmetry	Read only what the symmetry cannot change: exact (8.88e-16). A false symmetry is a wall.
Neural ODE	Learn the next-step rule, not the answer: 1.7 against 11.6 at three times the length.
Neural operator	A whole family at once: a fraction of a millisecond per new case instead of seconds; less accurate.
Diffusion	A known physical process run backwards, one small step at a time.
Choosing	Match the method to the knowledge. A standard equation on a simple shape wants a classical solver.

7.1 Twelve words, in plain English

law	something that is always true, exactly, whether or not you have data
residual	what is left over when a guess is put back into the equation; zero means it obeys
physics loss	the residual squared and averaged over many free checkpoints

collocation point	a place where the equation is checked; no measurement needed
derivative, slope	how fast something is changing
automatic differentiation	the computer working out slopes exactly by tracking every step
initial condition	where the spring starts and how fast it moves at time zero
soft penalty	a term in the loss asking for a condition, with a weight
hard constraint	the condition built into the model's arithmetic so it cannot be broken
conservation	a quantity that stays exactly constant over time
invariant model	a model that reads only what a symmetry cannot change
surrogate	a fast stand-in for a slow simulation

8 Twenty practice questions, with answers

COVERS Quiz preparation
TIME 20 minutes
IN BRIEF Cover the right-hand column, answer, uncover. Every question is one of the eight quiz topics in disguise.

#	Question	Answer
1	In one line each, what does a pattern do and what does a law do to the set of possible answers?	A pattern nudges towards likely answers and rules nothing out; a law removes a whole set of answers permanently.
2	How can you mark an answer to "find the number whose square is 9" without the answer sheet?	Put the answer back into the question: $4 \times 4 = 16 \neq 9$, so 4 is wrong. Checking needs no answer sheet.
3	A model that never saw a measurement scored 0.00135; one given 64 scored 0.026. What was different between them?	Only the check: distance from measurements against how far from obeying the equation. Same network, optimiser and steps.
4	Why does this matter for a bridge with forty sensor readings?	More data is not for sale; the only lever left is what you already know, and the physics of bridges is known.
5	What is a residual?	How far a guess is from obeying the equation: the leftover when you substitute it back. Zero means it obeys.
6	Why are collocation points free?	Checking an equation needs no measurement, so a checkpoint can be placed anywhere, even where measuring is impossible.
7	Is a physics-informed network a new kind of network?	No. It is an ordinary network with an unusual check; there is no PINN architecture.
8	What is the one new capability the method needs, and why is nudging the input not enough?	The slope of a slope, exactly. Nudging divides a tiny difference by a tiny number squared and rounding swamps it.
9	Both models had data on the first 40% of the time. What happened beyond it?	Data-only got 27.6 times worse; physics-informed 1.0, because the law was known out there too.
10	Why is "it does not know that it does not know" the dangerous property?	The data-only model fails smoothly and silently outside its data, so nothing warns you.
11	Energy was in nobody's loss. What happened, and what does it show?	Data-only drifted 16.3%, physics-only 0.2%: a model that obeys the equation inherits everything the equation implies.
12	How do you tell a memoriser from an understander?	Ask a question that was not on the practice sheet: check a consequence you never trained on.
13	Three soft-penalty runs all got the start right to four decimals and had errors of 0.0015, 0.32 and 0.66. What went wrong?	The weight λ was badly chosen; the bad runs obeyed the start while neglecting the equation. Nobody can derive λ.

14	What does a hard constraint change?	The start is built into the model's arithmetic (a fade-in that is zero at t = 0), so it holds automatically; one-part loss, no λ.
15	Ten checkpoints gave a smaller loss and a larger true error than 200. Why?	The loss reports obedience only at the places checked; between ten places the curve wanders. The loss answers a different question.
16	Told the friction is four times too high, how did the physics model compare with data-only, and would the loss have warned you?	Seven times worse (0.146 against 0.022), and no: the reported loss stayed small throughout.
17	Why does a model that reads pairwise distances change by 8.88e-16 when the input is turned?	Distances do not change under turning, so the model cannot see the turn; the number is rounding error.
18	With gravity added, the distance model scored 0.24 and could not improve. Why?	It was built unable to see which way up the arrangement is; a false symmetry is a wall at any dataset size.
19	What is the difference between a PINN and a neural ODE, in one line each?	A PINN learns one answer (one motion from one start); a neural ODE learns the rule for the next step and can start anywhere.
20	You have a standard equation on a simple rectangle. What should you use, and why?	A classical solver: faster, more accurate, with guarantees. Physics-informed learning is not a better way to solve standard equations.